

DULIPUDI LAASHMITH SANJAY

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SUMMARY

Electronics and Communication Engineering graduate with a strong foundation in Digital Electronics, CMOS fundamentals, and VLSI design principles. Hands-on experience in circuit design, simulation, and Verilog-based digital system development, complemented by practical exposure to industry-standard electronic design automation (EDA) tools. Demonstrates strong analytical thinking, problem-solving, attention to detail, and effective teamwork, and is seeking opportunities to contribute to the electronics and semiconductor industry.

EDUCATION

B.Tech - Electronics and Communication Engineering 2022 – 2026 | Guntur, India
Vasireddy Venkatadri Institute of Technology
CGPA: 8.35
Relevant Coursework: Digital & Analog Electronics, CMOS VLSI Design, Digital IC Design, Semiconductor Devices.

WORK EXPERIENCE

VLSI Design Intern 2025/05 – 2025/06 | Vijayawada
ChipIN Centre – C2S Programme, MeitY (Govt. of India)
- Gained hands-on experience across Analog IC Design, Digital RTL Design, and Physical Design workflows using Cadence Virtuoso, Synopsys VCS/DC/ICC2, and Vivado tools.
- Developed and verified Verilog RTL modules, including adders, multipliers, counters, and finite state machines (FSMs), with functional debugging through simulation and waveform analysis.
- Performed RTL synthesis, timing and area analysis, floorplanning, placement, clock tree synthesis (CTS), and routing as part of the digital physical design flow.

Vice President of IETE 2024/06 – 2025/05 | Guntur
The Institution of Electronics and Telecommunication Engineers
- Led the planning and execution of technical workshops, seminars, guest lectures, competitions, and student engagement initiatives in collaboration with faculty, industry professionals, and student volunteers.
- Managed chapter operations by coordinating cross-functional teams, promoting student participation through outreach activities, and strengthening leadership, communication, teamwork, and organizational skills.

PROJECTS

64-Tap FIR Filter (RTL Design, Synthesis & Verification) 2025/05 – 2025/06
- A digital signal processing project focused on implementing a modular FIR filter using Verilog HDL. Designed and implemented a parameterizable 64-tap FIR filter in Verilog RTL, using custom multipliers and adders to achieve high throughput and modularity.
- Synthesized RTL code into a gate-level netlist using Synopsys Design Compiler, applied timing constraints, and optimized trade-offs between speed and area utilization. Developed a self-checking Verilog testbench to verify functional correctness against a golden reference model, ensuring accuracy across multiple test cases.
- Performed detailed waveform analysis and signal tracing in Synopsys Verdi to debug and validate the design.

Two-Stage CMOS Op-Amp (Analog & Layout Design) 2025/05 – 2025/06
- Engineered a two-stage CMOS Op-Amp in Cadence Virtuoso, implementing a differential input stage with a current mirror load and a common-source stage with Miller compensation to achieve high gain and stability.
- Analyzed key performance metrics, including gain, bandwidth, slew rate, and phase margin, by conducting comprehensive AC, DC, and Transient simulations. Executed the complete physical design flow from schematic capture to final layout, performing post-layout simulations to validate design accuracy and performance.
- Validated the layout integrity by conducting rigorous Physical Verification, successfully resolving all Design Rule Check (DRC) and Layout vs. Schematic (LVS) violations. Optimized design trade-offs between low power consumption, stability, and a high gain-bandwidth product to meet performance targets.

SKILLS

Programming Languages — Verilog, C (Moderate), Python (Moderate),
Tools — Cadence Virtuoso, Synopsys Design Compiler, Synopsys Verdi, Xilinx Vivado, MATLAB/Simulink
Technical Skills — CMOS Circuit Analysis, Combinational & Sequential Logic, Circuit Design & Simulation, HDL Implementation, Low-Power Design Fundamentals, Basic Understanding of PD Concepts (Floorplanning, Placement, Routing)